

TASK 2 REPORT

TASK 3: PROTEIN STRUCTURE PREDICTION AND VISUALIZATION

1. Objective

The aim of this project was to predict the 3D structure of the Hemoglobin subunit beta protein and visualize it to understand its structural arrangement, specifically its helices and loops. This involves using homology modeling and visualization tools to interpret the molecular design.

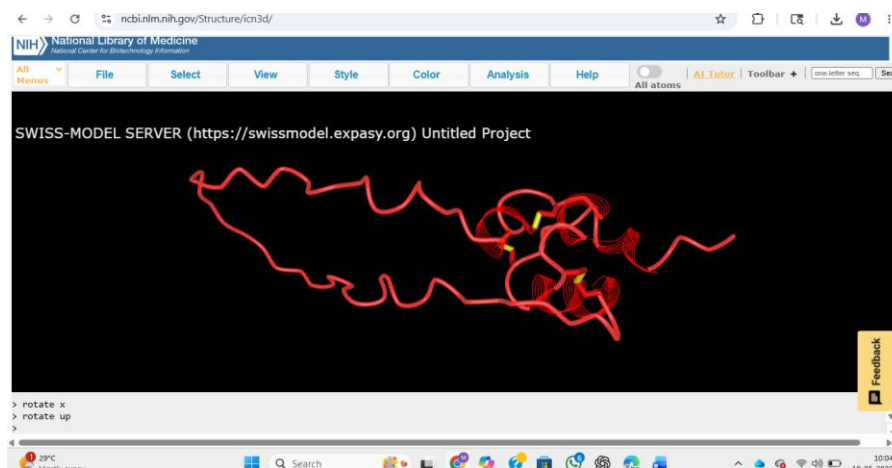
2. Methods and Tools Used

Step	Tool / Method
Sequence Retrieval	UniProt (Hemoglobin subunit beta)
Structure Prediction	SWISS-MODEL (Homology Modeling)
Visualization	iCn3D Web Viewer (NCBI)

3. Results & Observations

A. 3D Model Prediction

I used the SWISS-MODEL server to build a model for the protein. Based on the homology with known templates, the server generated a PDB file. The visualization shows a clear secondary structure.



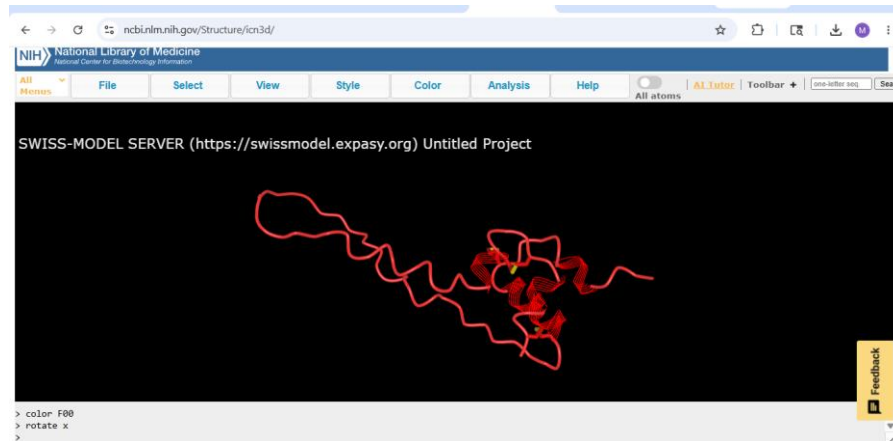
B. Structural Features

Upon loading the model in **iCn3D**, I changed the style to 'Ribbon' and color to 'Secondary Structure'. I observed the following:

- **Alpha Helices:** The protein is highly helical. In the magenta/red colored model, several spirals

represent these helices which are typical for globins.

- **Loop Regions:** There is a long unstructured region or loop visible on one side of the protein. This suggests a flexible tail or a part of the protein that does not have a rigid structure in the used template.
- **Ligand Binding:** The space within the helical clusters is where the Heme group binds in functional Hemoglobin.



4. Discussion

The visualization confirms that the protein belongs to the globin family. The prediction was successful, though the large disordered loop at the end (as seen in the screenshots) shows that the protein has some highly flexible regions. Using tools like iCn3D allowed me to rotate and inspect these coils from different angles to understand the folding.

5. Conclusion

The 3D structure prediction and visualization task was completed successfully. I have documented the structure through multiple screenshots and analyzed the helical nature of the Hemoglobin subunit beta protein.

